Cardiac Deformation Lab

Image-Based Cardiac Deformation Mapping & Kinematic Strain Analysis

Analysis ID:	9c3fc27e-3e93-4d18-ac85-b3b2346c94dd	**Analysis Type:**	Multi-Frame Temporal Sequence Analysis
Execution Date:	October 15, 2026 at 16:11:02 UTC	**Motion Model:**	Dense Farnebäck Optical Flow
Frames Analyzed:	2	**Consecutive Pairs:**	1

1. Displacement & Kinematic Motion

Metric Parameter	Value (Pixel Units)	Description
Mean Displacement	N/A	Average Euclidean magnitude of displacement vector field
Maximum Displacement	N/A	Peak regional apparent motion amplitude across the field
Minimum Displacement	N/A	Lowest regional motion magnitude observed
Displacement Std. Dev.	N/A	Spatial heterogeneity of tissue displacement

2. Small-Deformation Strain Tensor & Principal Fields

Strain Component	Mean	Min	Max	Std Dev
Normal Exx (du/dx)	N/A	N/A	N/A	N/A
Normal Eyy (dv/dy)	N/A	N/A	N/A	N/A
Shear Exy (0.5*(du/dy+dv/dx))	N/A	N/A	N/A	N/A
Principal Strain 1 (Major)	N/A	N/A	N/A	N/A
Principal Strain 2 (Minor)	N/A	N/A	N/A	N/A

3. Advanced Deformation Metrics & Geometric Projection

Metric Parameter	Observed Value	Physical / Mathematical Interpretation
Motion Coherence	N/A	Cosine consistency of motion vectors within local 5x5 spatial kernels
Strain Gradient Magnitude	N/A	Spatial rate of change of principal strain fields
Strain Anisotropy Index	N/A	Directional preference of tissue deformation (|e1 - e2| / (|e1| + |e2|))
Strain Entropy	N/A	Shannon entropy describing dispersion of regional strain histograms
Deformation Energy	N/A	Pseudo-elastic strain energy density surrogate (0.5 * Tr(eps * eps))

Radial Deformation (Mean)	N/A	Motion projection along radial axes from ventricular centroid approximation
Circumferential Deformation (Mean)	N/A	Motion projection orthogonal to radial vectors (tangential shortening)

4. Multi-Frame Temporal Strain Trajectory

Temporal Parameter	Value	Mathematical Definition
Temporal Mean Principal Strain	4.1490e-20	Average principal strain magnitude across all consecutive frame pairs
Temporal Std Dev	0.0000	Standard deviation of frame-to-frame strain trajectory
Trajectory Range (Min - Max)	4.1490e-20 to 4.1490e-20	Minimum and peak principal strain observed during cine sequence
Temporal Stability Index (TSI)	1.0000	Smoothness metric 1 / (1 + variance) measuring trajectory stability

5. Cardiac Deformation Fingerprint (10-Dimensional)

Dimension Symbol	Canonical Name	Value
D_mean	Mean Displacement	N/A
D_max	Maximum Displacement	N/A
epsilon_mean	Representative Mean Strain	N/A
epsilon_max	Representative Max Strain	N/A
epsilon_principal	Representative Principal Strain	N/A
AI	Strain Anisotropy Index	N/A
H	Strain Entropy	N/A
G_epsilon	Strain Gradient Magnitude	N/A
C	Motion Coherence	N/A
TSI	Temporal Stability Index	N/A

7. Technical Interpretation & Research Prototype Disclaimer

Methodological Scope: The metrics, vector fields, and spatial colormaps detailed in this report represent apparent image-space deformation computed via Farnebäck dense optical flow and classical continuum strain tensor decomposition. Measurements are expressed in image coordinate units (pixels) and relative mathematical differentials.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:
This software is an experimental computational research prototype. All values, including the Cardiac Deformation Fingerprint (10-D) and Temporal Stability Index (TSI), are algorithmic descriptors of pixel kinematics. This system does NOT provide medical diagnoses, clinical risk assessments, normal/abnormal classifications, or treatment recommendations. This report must NOT be used as a medical device or diagnostic decision-making instrument.